Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/04/2006
PI Name:	Hau, Lene V Ph.D.		
Project Title:	Ultra-Slow and Stopped Light in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hau@physics.harvard.edu	**Fax:**	FY 617-496-0189
PI Organization Type:	UNIVERSITY	**Phone:**	617-496-5967
Organization Name:	Harvard University		
PI Address 1:	Department of Physics		
PI Address 2:	17 Oxford Street		
PI Web Page:	http://www.deas.harvard.edu/		
City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2909		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have succeeded in slowing light pluses in a Bose-Einstein condensate (BEC) to only 17 m/s, more than seven orders of magnitude lower than the light speed in a vacuum. More recently, we completely stopped, storred, and retreived light pluses after storage times of several mimmiseconds. Associated with light speed reduction is an extreme spatial compression of pluses. In particular, with the most recent extension of our method, the 'light roadblock,' we have compressed light pluses from a length of 2 km to only 2ùm at the center of a condensate. At these length scales we can create extremely localized defects and generate dramatic excitations in the condensates. During the funding period we plan to perform ground-based experiments to pursue studies of controlled storage of optical information and ensuing nonlinear dynamics and coherent processing of the stored information in trapped superfluid Bose-Einstein condensates. Atom-atom interactions control the nonlinearities and processing dynamics during the storage time. The lowest light speeds--and thereby maximum spatial compression factors--obtainable in ground based experiments are limited by inhomogeneous effects due to the velocity spread in the atom clouds from finite temperature effects, and to Zeeman		

Task Description:	shifts from the magnetic trapping potential needed to keep the atoms in place. The maximum storage and processing times are also limited by finite temperature effects from incoherent smearing of the stored information and from dephasing and damping dure to the presence of non-condensed atoms surrounding the condensates. In a microgravity environment, we could cool atom clouds to much lower temperatures and could subsequently soften the magnetic trapping potential significantly, without causing sagging of the cloud due to gravity. We could produce condensates with macroscopic coherence lenghts, that could be trapped in the absence of non-condensed atoms. We could even turn off the trapping potential which would result in a very controlled atom cloud expansion, without a rapid acceleration and displacement of the cloud in gravity. For a carefully picked delay time after trap release, we could inject light pluses into a condensate of the exactly right spatial shape, a fact of extreme importance for studies of many-body systems of quantized vortices insuperfluid BECs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our recent work reveals collisions between tiny solitons and quantized vortex rings inside a Bose-Einstein condensate (BEC) of sodium atoms. By spatially engineering a slow-light medium comprising BEC and `coupling' laser, two injected counter-propagating probe light pulses are ultra-compressed to widths of a few microns (a factor of 100 million smaller than in free space) at respective `light roadblocks'. The ultra-compressed pulses are completely stopped at the light roadblocks, resulting in two very deep and narrow density defects inside the BEC. In effect, two narrow slices are removed from the condensate, pushing it far off equilibrium. The dynamics that follow lead to quantum shock waves, solitons, and quantized vortex rings. With this double roadblock setup, the solitons and vortex rings collide with one another in the center of the cloud. By allowing the BEC to expand we are able to control the timing of collisions between these nonlinear excitations and to thus generate novel, hybrid nonlinear excitations. The excitations are part soliton, part vortex ring. The vortex rings, embedded in solitonic shells, act as phantom propellers and cause the hybrid structures in the condensate to bend, like umbrellas turning inside out. They also enable the hybrid structures to merge to form closed, low-density ellipsoidal shells, closing off and isolating tiny islands of superfluid from the rest of the world. The structures were imaged with a 3D tomography technique in which the central cross-section of the cloud is pumped to a different internal state where it can be selectively imaged. The resulting superfluid excitations demonstrate the degree of control that we have over BECs dynamics. The work was reported in Physical Review Letters, and highlighted in our invited paper in Optics and Phontonics News in May 2005 (this issue of Optics and Photonics News was handed out by the Optical Society of America at the major CLEO meeting in Baltimore). Our results were also publicized in the American Institute of Physics' Physics News Update, and in Physics Today's Physics Update. The paper was picked as one of the main news stories in physics in 2005 as it was selected for "Physics News in 2005" that appeared in the February 2006 issue of APS News. Under the NASA project we have also created the highest-flux source of cold atoms ever made. The work was published in the October 2006 issue of Review of Scientific Instruments (vol. 76, page 103101). This atom source could form the basis for a CW atom laser which is considered a holy grail in atomic physics and expected to be of immense importance as a source for state-of-the-art navigational gyroscopes based on atom interferometry. SUPPLEMENTAL INFORMATION FOR FINAL REPORT Optical information transfer and processing in Bose-Einstein condensates: During the final funding period, we have succeeded in demonstrating coherent optical information processing with slow and stopped light in Bose-Einstein condensates. In these experiments (see Nature 2007 citation in Bibliography), we stop and extinguish a light pulse in one Bose-Einstein condensate (BEC) and revive the light pulse in a totally separate condensate 160 microns away. The light pulse is converted to a traveling matter copy that is isolated in space, and hence can be trapped, manipulated and sent on to be revived as light in a second condensate. The work represents a new paradigm in quantum control, coherent optical information processing, and wavefunction sculpting. Selected Publicity: The work described above was published as the cover story of Nature, February 8, 2007, and was widely reported in the press in this country as well as throughout the world (Europe, Australia, South America, Indonesia). As two examples of the press coverage, the work was reported in the New York Times on the day of publication, and the PI appeared on National Public Radio in the directly transmitted program ‘Talk of the Nation: Science Friday’ (on February 9). An interview with the PI and recordings from the lab will be featured in the BBC/PBS special, Absolute Zero, that will air in 2007. A Swedish science program on our research was aired on the main Swedish TV channel on December 4, 2005, and will be rerun this year on another Swedish national channel “Kunskapskanalen”. The PI’s work on Stopping Light was recently high-lighted as one of a select few milestones in the 160 year history of the Harvard Applied Sciences and Engineering. Lene Hau was selected by the MacArthur Foundation as one of only 9 MacArthur Fellows to be featured in connection with the 25th anniversary of the MacArthur Fellows program. The 9 fellows were selected among all MacArthur fellows named over the history of the fellows program. See "MacArthur Fellows: the first 25 years, 1981-2005." (Published by the John D. and Catherine T. MacArthur Foundation, 2005). The Library of Congress has recognized Lene Hau in its "Women Who Dare" 2007 desk calendar of accomplished women from around the world and throughout history. http://pomegranate.stores.yahoo.net/
Bibliography Type:	Description: (Last Updated: 12/20/2019)
Abstracts for Journals and Proceedings	Slowe C, Ginsberg N, Ristroph T, Goodsell A, Hau LV. "Ultra-slow light and Bose-Einstein condensates: Two-way control with coherent light and atom fields." The May 2005 issue of Optics and Photonics News was distributed on site at the major Optical Society of America’s /CLEO /meeting in Baltimore. Optics and Photonics News, Optical Society of America, Invited Paper, May 2005. , May-2005

Articles in Peer-reviewed Journals	Ginsberg NS, Garner SR, Hau LV. "Coherent control of optical information with matter wave dynamics." Nature. 2007 Feb 8;445(7128):623-6. PMID: 17287804 , Feb-2007
Articles in Peer-reviewed Journals	Ginsberg NS, Brand J, Hau LV. "Observation of hybrid soliton vortex-ring structures in Bose-Einstein condensates." Phys Rev Lett. 2005 Feb 4;94(4):040403. Epub 2005 Jan 31. PMID: 15783535 , Feb-2005
Articles in Peer-reviewed Journals	Slowe C, Vernac L, Hau LV. "High flux source of cold rubidium atoms." Review of Scientific Instruments 2005 Oct;76(10):103101. http://dx.doi.org/10.1063/1.2069651 , Oct-2005
Articles in Peer-reviewed Journals	Soljacic M, Lidorikis E, Joannopoulos JD, Hau LV. "Ultralow-power all-optical switching." Applied Physics Letters 2005 Apr 25;86)17):171101. http://dx.doi.org/10.1063/1.1900956 , Apr-2005
Awards	Hau LV. "Selected by MacArthur Foundation as one of 9 MacArthur Fellow featured in 25th anniversary of MacArthur Fellow program, “MacArthur Fellows: the first 25 years, 1981-2005." Sep-2005
Journal/Magazine covers	Ginsberg NS, Garner SR, Hau LV. "Cover in journal Nature for article, “Coherent control of optical information with matter wave dynamics." Nature. 2007 Feb 8;445(7128):623-6. PMID: 17287804 , Feb-2007
Significant Media Coverage	Ginsberg NS, Garner SR, Hau LV. "Swedish science program summary of Hau et al. research." Swedish TV channel, Science Program, December 4, 2005., Dec-2005
Significant Media Coverage	Hau LV. "Appearance of PI and interview regarding research reported in Nature for “Coherent control of optical information with matter wave dynamics.“" National Public Radio's "Talk of the Nation: Science Friday" program, February 9, 2007., Feb-2007
Significant Media Coverage	Slowe C, Ginsberg N, Ristroph T, Goodsell A, Hau LV. "The May 2005 issue of Optics and Photonics News was distributed on site at the major Optical Society of America’s /CLEO /meeting in Baltimore, for invited paper, " The May 2005 issue of Optics and Photonics News was distributed at Optical Society of America’s /CLEO /meeting, May 2005., May-2005